Question:

Write the below text in the file called sample.txt and then find the frequency count of the patterns ‘pe’, and ‘pi’

Peter Piper picked a peck of pickled peppers

A peck of pickled peppers Peter Piper picked

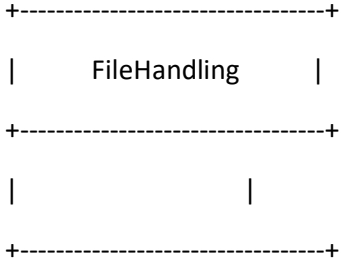
If Peter Piper picked a peck of pickled peppers

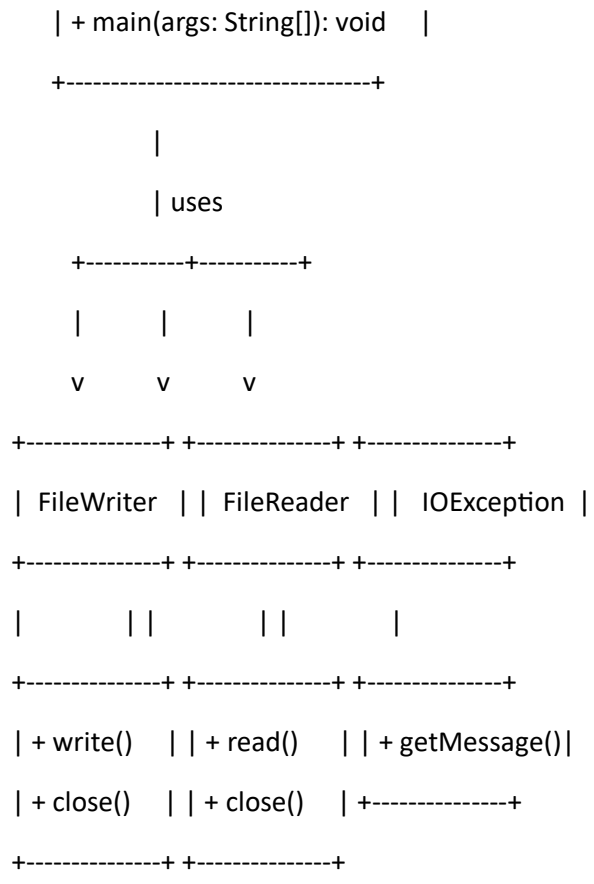
Where’s the peck of pickled peppers Peter Piper picked?

Algorithm:

1. Start
2. Create a string containing the given text.
3. Open sample.txt using FileWriter.
4. Write the text into the file and close the file.
5. Open sample.txt using FileReader.
6. Read the file character by character until the end of the file.
7. Store the read characters in a string named content.
8. Close the file reader.
9. Initialize peCount = 0 and piCount = 0.
10. Traverse the content and extract every two consecutive characters.
11. If the pattern is "pe" (case-insensitive), increment peCount.
12. If the pattern is "pi" (case-insensitive), increment piCount.
13. Display the number of occurrences of "pe" and "pi".
14. If an IOException occurs, display the error message.
15. Stop.

UML Class Diagram:





Code:

```
package javacore;

import java.io.FileWriter;
import java.io.FileReader;
import java.io.IOException;

public class FileHandling {
   public static void main(String[] args) {

      // Text to be written into the file
      String text = "Peter Piper picked a peck of pickled peppers\n"
            + "A peck of pickled peppers Peter Piper picked\n"
            + "If Peter Piper picked a peck of pickled peppers\n"
            + "Where's the peck of pickled peppers Peter Piper picked?";
```

```
try {
   // Create the file and write the text
   FileWriter fw = new FileWriter("sample.txt");
   fw.write(text);
   fw.close();

   // Open the file for reading
   FileReader fr = new FileReader("sample.txt");

   int ch;
   String content = "";

   // Read the file character by character
   while ((ch = fr.read()) != -1) {
      content = content + (char) ch;
   }

   // Close the file
   fr.close();

   // Initialize pattern counters
   int peCount = 0;
   int piCount = 0;

   // Check every two-character pattern
   for (int i = 0; i < content.length() - 1; i++) {

      // Extract two consecutive characters
      String pattern = content.substring(i, i + 2);
```

```
            // Count occurrences of "pe"
            if (pattern.equalsIgnoreCase("pe")) {
                peCount++;
            }

            // Count occurrences of "pi"
            if (pattern.equalsIgnoreCase("pi")) {
                piCount++;
            }
        }

        // Display the frequency counts
        System.out.println("'pe' - no of occurrences - " + peCount);
        System.out.println("'pi' - no of occurrences - " + piCount);

    } catch (IOException e) {
        // Handle file-related errors
        System.out.println("File error: " + e.getMessage());
    }
  }
}
```

Output:

'pe' - no of occurrences - 11

'pi' - no of occurrences - 4